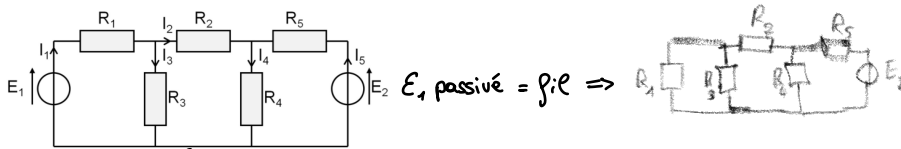


1 PARTIE 1 : CIRCUIT ELECTRIQUE DE VOILIER (INDEPENDANTE)

Question 1.



$$R_1 // R_3 : R_{13} = \frac{R_1 R_3}{R_1 + R_3} = \frac{R \times 1000R}{R + 1000R} = \frac{1000R^2}{1001R} = 1R = R$$

$$R_{13} - R_2 : R_{132} = R_{13} + R_2 = R + 100R = 101R$$

$$R_{132} // R_4 : R_{1324} = \frac{R_{132} \times R_4}{R_{132} + R_4} = \frac{101R \times 30R}{101R + 30R} = \frac{3030R^2}{131R} = \frac{3030}{131}R$$

$$R_{1324} - R_5 : R_{13245} = R_{1324} + R_5 = \frac{3030}{131}R + 100R = \frac{3130}{131}R$$

$$R_{eq} = \frac{3130}{131}R \checkmark$$

loi d'Ohm $\rightarrow I_5 = \frac{E_2}{\frac{3130R}{131}} = \frac{131E_2}{3130R} \checkmark$ loi des mailles $\rightarrow U_4 = E_2 - U_5 = E_2 - \frac{10}{313}E_2 = \frac{303E_2 - 10E_2}{313} = \frac{303}{313}E_2 \checkmark$

loi d'Ohm $\rightarrow U_5 = R_5 I_5 = \frac{100R}{131} \times \frac{131E_2}{3130R} = \frac{100}{3130}E_2 = \frac{10}{313}E_2 \checkmark$ $I_4 = \frac{U_4}{R_4} = \frac{\frac{303}{313}E_2}{30R} = \frac{303E_2}{313 \times 30R} = \frac{303E_2}{9390R} \checkmark$

Question 2.

$$I_4 = \frac{303 \times 12}{9390 \times 0,05} = 7,5 A$$

$$I_5 = \frac{131 \times 12}{3130 \times 0,05} = 10 A$$

Question 3.

Question 4.

Question 5.

2 PARTIE 2 : ANALYSE DE LA PROPULSION DU VOILIER

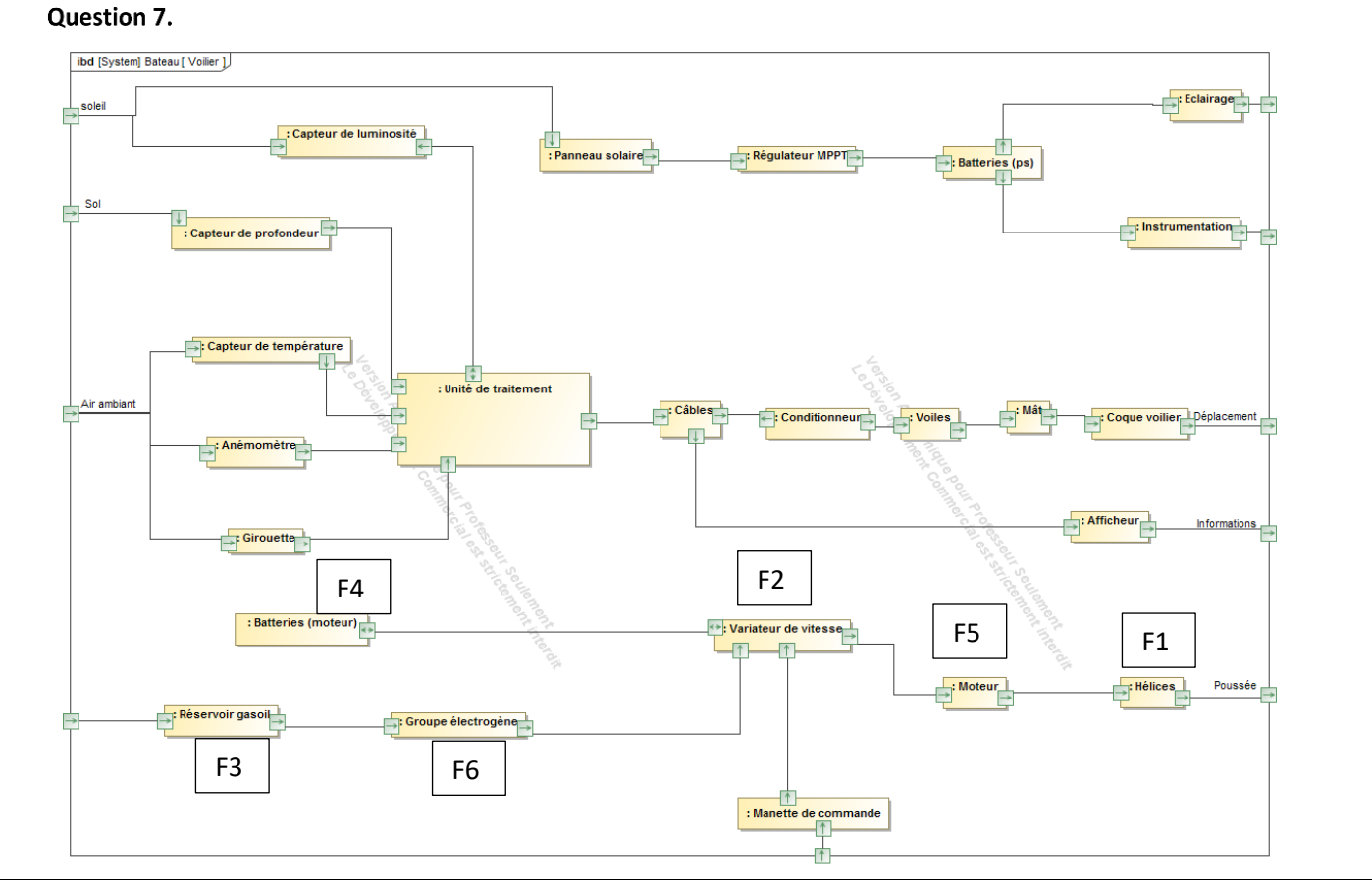
Question 6.

Raison :

Exigences d'autonomies :

-

-



Question 8.

F1		F4	
F2		F5	
F3		F6	

Question 9.

Question 10.

Question 11.

Question 12.

Question 13.

Question 14.

Question 15.

Question 16.

3 PARTIE 3 : ÉTUDE DU SYSTÈME DE QUILLE PENDULAIRE

Question 17.

ACQUERIR

Inclinomètre

Question 18.

Moment en P de la force vélique :

$$\overrightarrow{M(P, \vec{F}_v)} =$$

Moment en P du poids de la coque :

$$\overrightarrow{M(P, \vec{P}_c)} =$$

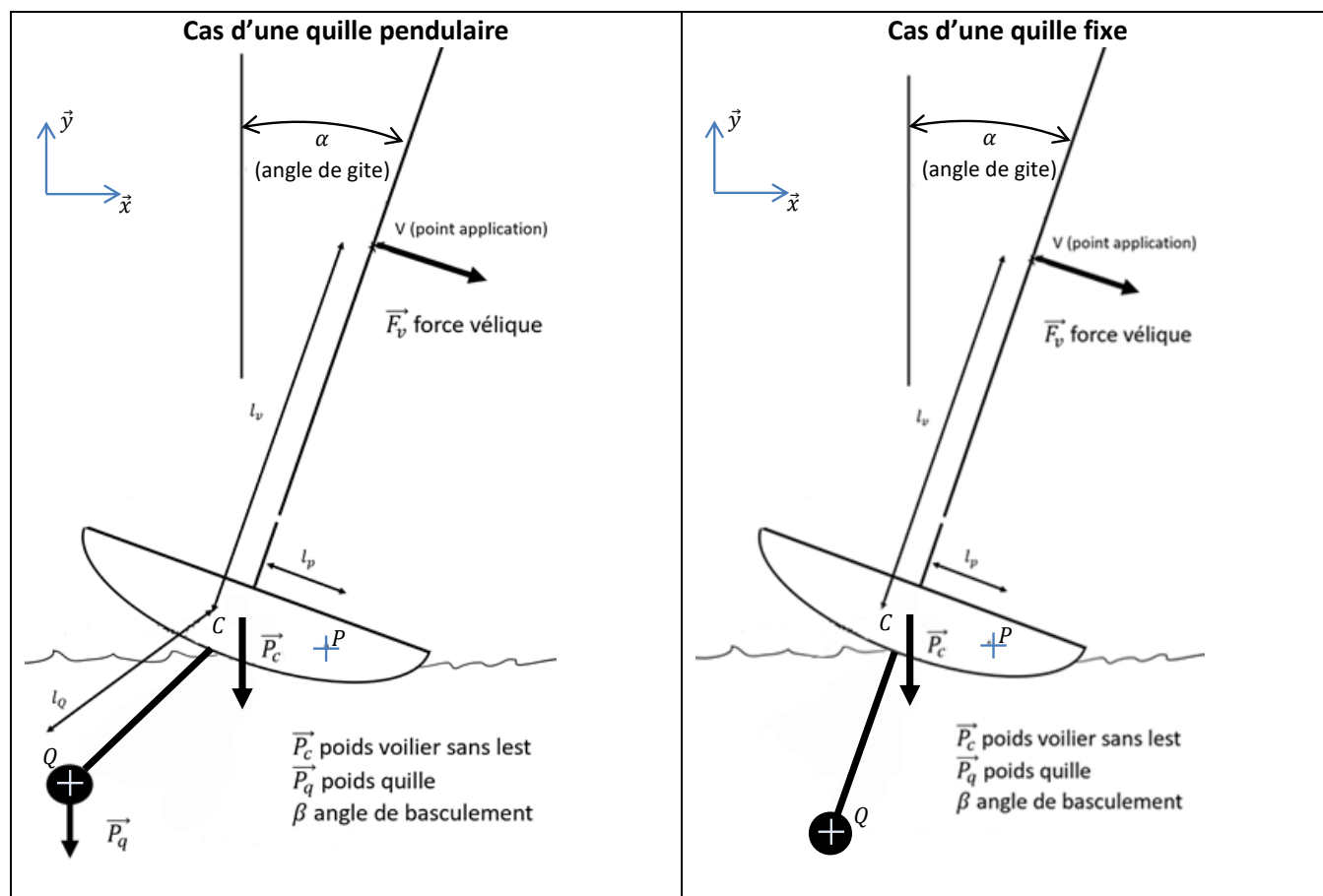
Moment en P du poids de la quille :

$$\overrightarrow{M(P, \vec{P}_q)} =$$

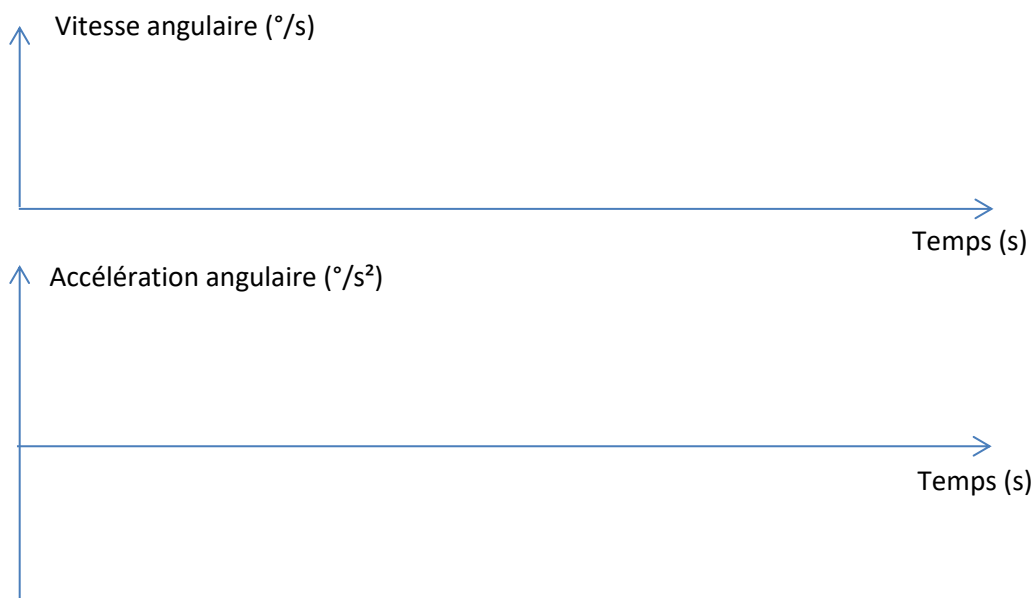
Question 19.

Question 20.

Question 21.



Question 22.



Question 23.

Question 24.

Question 25.

Question 26.

Question 27.

Question 28.

4 PARTIE 3 : ETUDE DES CHAINES D'ACQUISITION

NOM :

Question 29.

Question 30.

Question 31.

Question 32.

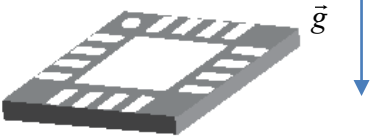
Question 33.

Question 34.

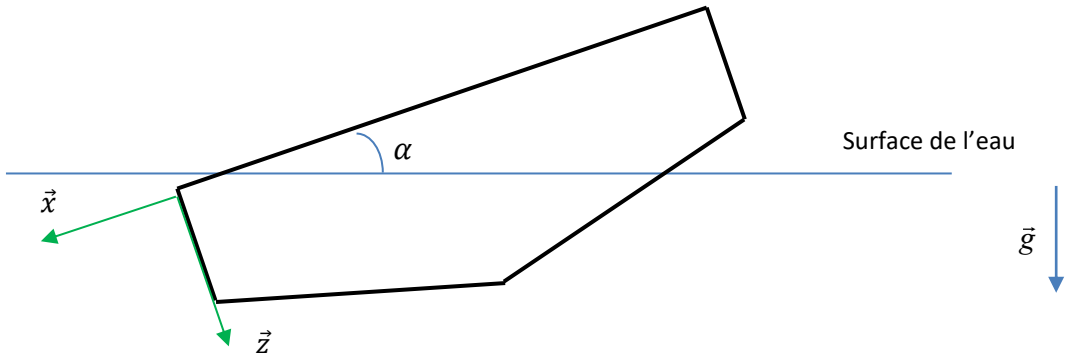
Question 35.

Question 36.

Question 37.

Situations	Sortie en g	Sortie en volt
	$X_{out} = 0$ $Y_{out} = 0$ $Z_{out} = 1$	$V_X =$ $V_Y =$ $V_Z =$
	$X_{out} = 0$ $Y_{out} = 0$ $Z_{out} = -1$	$V_X =$ $V_Y =$ $V_Z =$
	$X_{out} =$ $Y_{out} =$ $Z_{out} =$	$V_X =$ $V_Y =$ $V_Z =$
	$X_{out} =$ $Y_{out} =$ $Z_{out} =$	$V_X =$ $V_Y =$ $V_Z =$

Question 38.



Alors

Question 39.

Question 40.

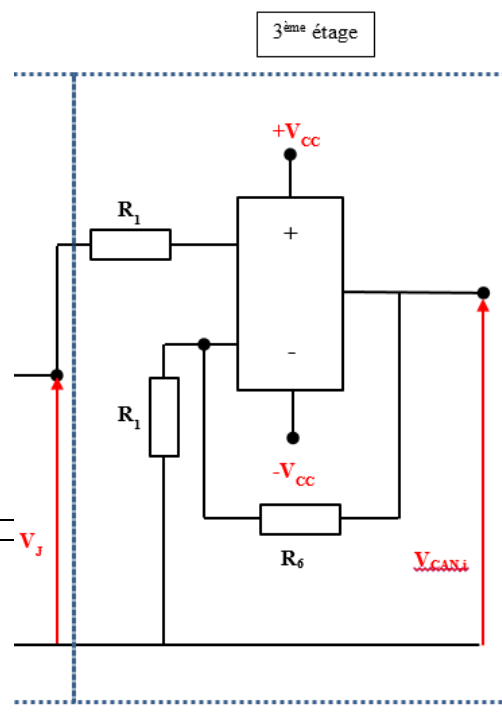
Question 41.

Question 42.

Question 43.

Question 44.

Question 45.



Question 46.

Question 47.